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Multi-Modal masked autoencoder and parallel Mamba for 3D brain tumor segmentation ☆

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ABSTRACT

Accurate segmentation of brain tumors from multimodal MRI is essential for diagnosis and treatment planning. However, most existing approaches can only process single type of data modality, without exploiting the complementary information across different modalities. To overcome this limitation, a novel framework called MFMamba which integrates modality-aware masked autoencoder pretraining, a gated fusion strategy, and a Mamba-based backbone for efficient long-range modeling is proposed. In this design, one modality is fully masked while others are partially masked, forcing the network to reconstruct missing data through cross-modal learning. The gated fusion module then selectively incorporates generative priors into task-specific features, enhancing multimodal representations. Experimental results on the BraTS 2023 dataset show that MFMamba achieves Dice score of 93.77% for Whole Tumor and 92.69% for Tumor Core, corresponding to 1.6–2.1% improvements over state-of-the-art baselines. The gains are statistically significant ($p < 0.05$), indicating the framework's ability to deliver more precise tumor boundary delineation. Overall, the results suggest that modality-aware fusion can enhance segmentation quality while maintaining computational efficiency, underscoring its potential application for clinical image analysis. The implementation is publicly available at https://github.com/ministerhuang/MFMamba.

1. Introduction

Accurate segmentation of brain tumors in three-dimensional magnetic resonance imaging (MRI) is essential for diagnosis, treatment planning, and prognosis evaluation. Multimodal MRI, including T1, T1-Gd, T2, and FLAIR sequences, provides complementary information that enables precise delineation of tumor subregions. The integration of these modalities, however, remains challenging due to variations in acquisition protocols, missing sequences, and heterogeneous tumor appearances.

Traditional convolutional neural networks (CNNs) such as U-Net [1], V-Net [2] have shown strong ability to capture local structures, yet their limited receptive fields hinder the modeling of irregular and global tumor boundaries. Transformer-based methods extend the receptive field through self-attention but incur high computational costs, restricting scalability to 3D volumes. Recently, state space models (SSMs), particularly Mamba and its segmentation-oriented variant SegMamba [3], have demonstrated efficient long-range modeling with linear-time complexity, making them promising backbones for volumetric medical imaging. However, existing SSM-based methods generally stack multimodal inputs without explicitly modeling modality-specific interactions, limiting their capacity to fully leverage complementary information.

In parallel, masked autoencoders (MAEs) [4] have emerged as powerful self-supervised learners. By reconstructing missing regions, MAEs capture semantic priors that are transferable to downstream tasks. In the medical domain, M3AE [5] has shown the potential of MAEs for handling missing-modality scenarios in brain tumor segmentation. Yet, the integration of modality-aware generative priors into end-to-end segmentation pipelines is still underexplored.

Beyond the field of neuroimaging, cross-domain studies provide additional evidence for the importance of multimodal fusion and efficient architectures. For example, semantic segmentation has been applied to microbial alterations using SegFormer backbones [6], while YOLO-based segmentation frameworks have been deployed for livestock monitoring [7] and crop disease detection [8]. Attention-enhanced deep vision models have improved food quality assessment [9], and

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blockchain-integrated deep learning frameworks have enabled secure medical classification in lung and colon cancer [10]. Further, machine learning has been used for hepatocellular carcinoma prediction [11], mental health monitoring [12], and other cross-domain applications such as super-resolution [13] and entropy-based edge detection [14]. Although these works target different problems, they collectively demonstrate that explicitly modeling cross-modal interactions and incorporating robust priors can lead to more reliable and generalizable systems.

To address above-mentioned challenges, we propose **MFMamba**, a modality-aware framework that integrates masked autoencoder pretraining with efficient SSM-based segmentation. Our main contributions can be summarized as follows:

- We introduce a modality-aware masking technique for generative pretraining, in which one modality is fully masked while others are partially masked. The design enables the network to reconstruct the missing modality, thus enhancing cross-modal representation learning.
- We design a gated fusion module that selectively merges generative priors with task-driven features, enabling more effective and reliable multimodal integration.
- We implement a Mamba-based segmentation backbone with modality-aware fusion, which achieves a balance between computational efficiency and segmentation accuracy for 3D brain tumor analysis.

Extensive experiments on the BraTS 2023 dataset demonstrate that the proposed MFMamba framework achieves consistent improvements over strong baselines, underscoring the value of modality-aware generative priors for robust multimodal MRI segmentation.

2. Related work

Brain Tumor Segmentation. Deep learning has advanced brain tumor segmentation from multimodal MRI. Convolutional neural networks (CNNs), such as U-Net [1], V-Net [2], and 3D U-Net [15], have demonstrated strong performance in capturing local structures but remain constrained by their limited receptive fields, reducing their effectiveness in modeling complex tumor boundaries. To overcome this, Transformer-based architectures have been proposed, including UNETR [16] and Swin-UNETR [17], which employ global self-attention to extend receptive fields. However, the quadratic complexity of attention mechanisms substantially increases computational costs, limiting their scalability to high-resolution volumetric data. Recent developments in state space models (SSMs), such as Mamba [18] and SegMamba [3], have emerged as efficient alternatives for long-range dependency modeling. These methods achieve linear-time sequence modeling while preserving representational power, making them a more computationally feasible solution. Nonetheless, existing SSM-based approaches often process multimodal MRI inputs through simple stacking, without explicitly capturing modality-specific interactions. In parallel, CAD systems for brain tumors have been developed to combine multimodal MRI with robust classification pipelines, further illustrating the importance of integrating modality-aware strategies [19].

Masked Autoencoders and Representation Learning. Masked autoencoders (MAEs) [4] have emerged as powerful self-supervised learning tools by reconstructing missing parts of the input data. They have been widely applied in medical imaging, where incomplete acquisitions are common. M3AE [5] extends this concept to the challenge of missing modalities in brain tumor segmentation, showcasing the ability of MAEs to handle heterogeneous inputs. More recently, multimodal MAEs [20] have been proposed to improve cross-modal alignment and consistency, further enhancing the utility of generative pretraining. These studies highlight the potential of modality-aware reconstruction for robust representation learning. Despite these advances, the integration of such generative priors into end-to-end segmentation frameworks is still limited.

Medical and Multimodal Applications. Multimodal and feature-driven methods have also been explored in diverse biomedical tasks, including brain lesion boundary refinement, multimodal diagnostic modeling, and the design of efficient architectures for medical image segmentation [21–23]. Beyond healthcare, related approaches have been applied to areas such as cancer prediction, microbial analysis, face recognition, and classical computer vision problems including super-resolution, edge detection, and content filtering [24–27]. Although these studies differ from brain tumor segmentation, they collectively highlight the broad applicability of multimodal learning strategies.

In summary, prior research reveals three persistent gaps: (i) CNN and Transformer-based methods either lack efficient long-range modeling or face high computational demands; (ii) while SSM-based methods such as SegMamba improve scalability, they typically ignore modality-aware fusion; and (iii) MAE-based approaches have demonstrated strong potential for cross-modal representation learning but remain underexplored in end-to-end multimodal segmentation pipelines. The framework proposed in this study, MFMamba, addresses these gaps by combining modality-aware MAE pretraining, a gated fusion mechanism, and an efficient Mamba-based backbone.

3. Method

We propose MFMamba, a novel framework for multimodal brain tumor segmentation that integrates generative pretraining with long-range dependency modeling. An overview of the MFMamba architecture is shown in Fig. 1. MFMamba consists of two components: a modality-aware masked autoencoder (MAE) and a Mamba-based segmentation network with a Gated Fusion Feature (GFF) module. During pretraining (upper part of Fig. 1), the MAE masks one modality completely and partially masks the others, driving the model to reconstruct the fully masked modality using the remaining modalities. This cross-modal reconstruction helps the network learn shared representations that capture complementary information.

After pretraining, the segmentation phase begins (lower part of Fig. 1). The GFF module fuses the features from the masked modalities with the learned priors. The unmasked modalities are processed normally, while the masked ones benefit from the prior information, enriching their feature representations. These features are then passed through the Mamba-based segmentation network, which models long-range dependencies across depth, height, and width, enabling accurate tumor segmentation. The model is jointly optimized with both the MAE reconstruction loss and segmentation loss, ensuring the priors contribute to the segmentation task.

3.1. Problem formulation

Brain tumor segmentation with multimodal MRI is a voxel-level multi-class problem. Given input volumes $\mathbf{X} = \{X^1, \ldots, X^M\}$ from modalities such as T1, T1-Gd, T2, and FLAIR, the goal is to predict a label map $\mathbf{Y} \in \{0, 1, \ldots, C-1\}^{H \times W \times D}$ covering whole tumor (WT), tumor core (TC), and enhancing tumor (ET). Each modality provides complementary cues (e.g., T1 for anatomy, T1-Gd for enhancement, T2 for edema, FLAIR for fluid-suppressed lesions). The segmentation network learns

$$f_\theta : \mathbf{X} \rightarrow \mathbf{Y}, \tag{1}$$

with parameters θ optimized by a supervised segmentation loss $\mathcal{L}_{\text{seg}}$ over labeled data.

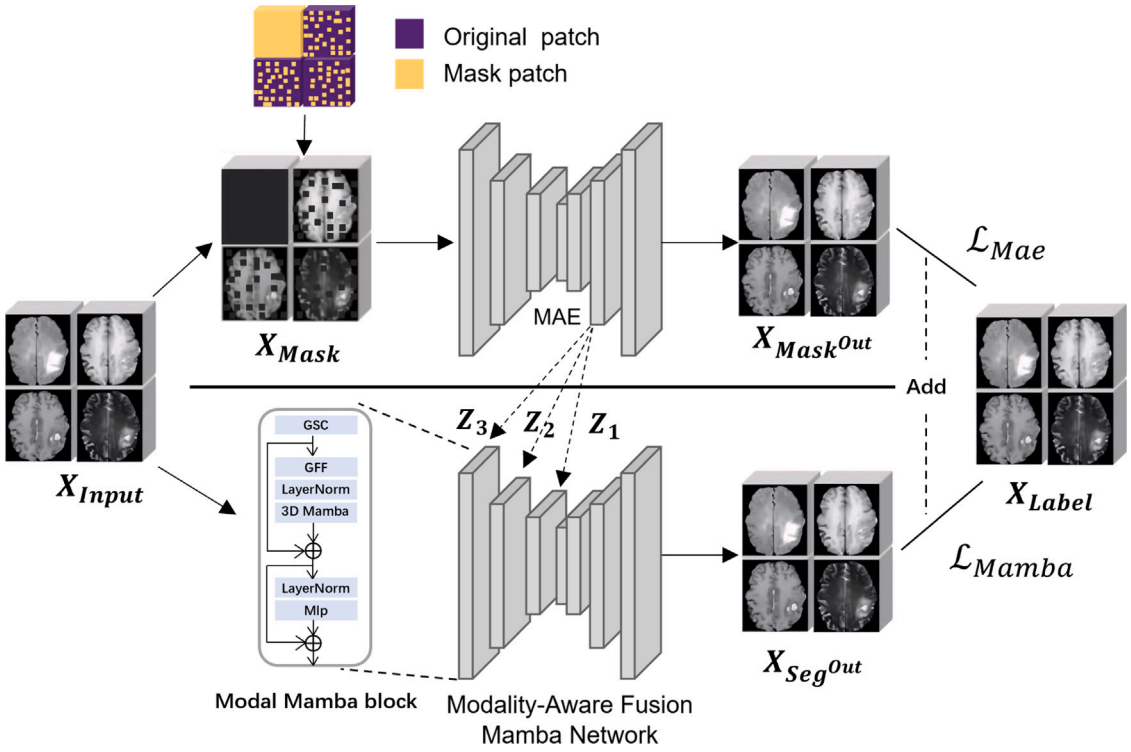


Fig. 1. Overview of the MFMamba model. It includes two stages: pretraining (top) where the network reconstructs masked modalities, and segmentation (bottom) where a Mamba-based network performs tumor segmentation. The losses L_{Mae} and L_{Mamba} are used in pretraining and segmentation, respectively. X_{Mask} and X_{MaskOut} refer to the input with masked regions and the reconstructed output, respectively. X_{SegOut} is the final segmented output, and Z_1, Z_2, Z_3 are intermediate features from the Mamba network.

3.2. Modality-aware masked pretraining

The MAE architecture used in MFMamba follows a typical encoder–decoder structure designed for 3D medical image segmentation. The encoder consists of four convolutional blocks, each containing three 3D convolutional layers, progressively downsampling the input while increasing the number of channels from 16 to 128. The decoder, similarly, consists of three upsampling and convolutional layers that restore the spatial resolution of the image and use skip connections to combine features from the encoder. This structure enables the model to effectively capture both global and local features, facilitating accurate modality reconstruction and improving the model's cross-modal inference capabilities, especially when handling missing modalities during segmentation [4].

To exploit cross-modal dependencies, we adopt a modality-aware masking strategy: in each iteration, one modality is fully masked while the others are partially masked with ratio $\rho = 0.75$. Formally,

$$\tilde{\mathbf{X}} = \left\{ \mathcal{M}_1(X^{m'}),\ \mathcal{M}_\rho(X^m),\ m \neq m' \right\}, \tag{2}$$

where $\mathcal{M}_1$ removes all patches and $\mathcal{M}_\rho$ randomly masks a fraction ρ of patches.

The pretraining objective is to reconstruct each modality from $\tilde{\mathbf{X}}$:

$$\mathcal{L}_{\text{MAE}} = \frac{1}{M} \sum_{m=1}^{M} \left\| X^m - \hat{X}^m(\tilde{\mathbf{X}}) \right\|_2^2. \tag{3}$$

This design compels the model to infer a fully missing modality from the remaining ones while denoising partially masked inputs, yielding stronger generative priors for downstream segmentation.

3.3. Gated fusion feature module (GFF)

To integrate generative priors into segmentation, we propose a Gated Fusion Feature (GFF) module that adaptively merges the features of masked modalities with MAE-generated guidance.

For a masked modality m, its feature X_m and the corresponding prior X_{MAE} are concatenated and passed through a lightweight gating network:

$$G = \sigma\left(W_2 * \text{ReLU}(W_1 * [X_m; X_{\text{MAE}}])\right), \tag{4}$$

where W_1 and W_2 are $1 \times 1 \times 1$ convolutions and $\sigma(\cdot)$ is the sigmoid function.

The fused representation is then computed as:

$$F_m = G \odot X_m + (1 - G) \odot X_{\text{MAE}}. \tag{5}$$

Unmasked modalities remain unchanged, and the final multimodal representation is obtained by concatenating all feature maps. This mechanism enables masked modalities to benefit from generative priors without distorting unmasked features, improving multimodal integration and segmentation accuracy.

3.4. Mamba segmentation network with modality-aware fusion

Our segmentation backbone builds upon SegMamba [3], which employs a tri-orientated Mamba (ToM) module to capture long-range dependencies along depth, height, and width, and a Gated Spatial Convolution (GSC) module for local refinement. While effective, SegMamba processes all modalities uniformly, neglecting their distinct and complementary characteristics.

To address this, we introduce a modality-aware extension. Each modality first passes through a GSC layer. For masked modalities, their features are further enriched using the Gated Fusion Feature (GFF) module, which incorporates generative priors from MAE pretraining:

$$Z_m = \begin{cases} \text{GFF}(\text{GSC}(X_m), X_{\text{MAE}}), & \text{if } m \text{ masked} \\ \text{GSC}(X_m), & \text{otherwise.} \end{cases} \tag{6}$$

Algorithm 1 Training Process for MFMamba

Require Dataset D, Epochs E, Learning rate η
Ensure Trained model parameters θ
for epoch = 1 to E **do**
  for each batch (X, Y) in D **do**
    Step 1: MAE Pretraining
    Randomly mask a modality $X^{m'}$ and partially mask others
    Reconstruct the masked modality using cross-modal learning
    Compute MAE loss $\mathcal{L}_{\text{MAE}}$
    Step 2: Gated Fusion Feature (GFF)
    for each modality m **do**
      if modality m was masked during pretraining **then**
        Apply GFF: $F_m = G \odot X_m + (1 - G) \odot X_{\text{MAE}}$
      else
        Proceed with original modality X_m
      end if
    end for
    Step 3: Mamba Segmentation
    Pass fused features F_m through Mamba network
    Compute segmentation loss $\mathcal{L}_{\text{seg}}$
    Step 4: Total Loss
    Update parameters $\theta \leftarrow \theta - \eta \nabla_\theta (\mathcal{L}_{\text{seg}} + \lambda \mathcal{L}_{\text{MAE}})$
  end for
end for

The fused representations are then processed by Mamba modules for global dependency modeling across spatial axes:

$$Z_{\text{out}} = \sum_{a \in \{d,h,w\}} \text{Mamba}(Z_a). \tag{7}$$

Finally, outputs from all modalities are aggregated into a unified representation for segmentation. This design ensures that masked modalities benefit from generative priors while unmasked ones retain their original structure, enabling the model to fully exploit cross-modal complementarities.

3.5. Loss function

To ensure that generative priors learned during pretraining contribute meaningfully to the segmentation task, we adopt a joint training objective. This combines the loss from the segmentation network with the reconstruction loss from the MAE phase. The total loss function is defined as:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{seg}} \mathcal{L}_{\text{seg}} + \lambda_{\text{MAE}} \mathcal{L}_{\text{MAE}} \tag{8}$$

where $\mathcal{L}_{\text{seg}}$ is the segmentation loss, $\mathcal{L}_{\text{MAE}}$ is the MAE reconstruction loss, and $\lambda_{\text{seg}}, \lambda_{\text{MAE}}$ are weighting coefficients. Through this combined objective, the segmentation network is encouraged to incorporate knowledge from generative pretraining, leading to improved performance. The training procedure of the proposed MFMamba model is given in Algorithm 1.

4. Experiments

In this section, we present experimental evaluation of the proposed MFMamba framework for brain tumor segmentation using 3D MRI.

4.1. Datasets

We used the BraTS 2023 dataset [28] which includes four MRI modalities per case: T1, T1-Gd, T2, and FLAIR. The dataset contains 1,251 cases, 875 of which are annotated with segmentation labels for training and validation, and 376 are reserved for testing with hidden ground truth. We split the labeled dataset into 70% training, 20% validation, and 10% internal testing (See Table 2). All scans are skull-stripped, aligned to a standard anatomical template, and resampled to a 1 mm isotropic resolution. The four modalities are stacked as input channels for the segmentation network. In addition, standard data augmentation techniques such as random flipping, rotation, and intensity normalization were applied during training to enhance model generalization.

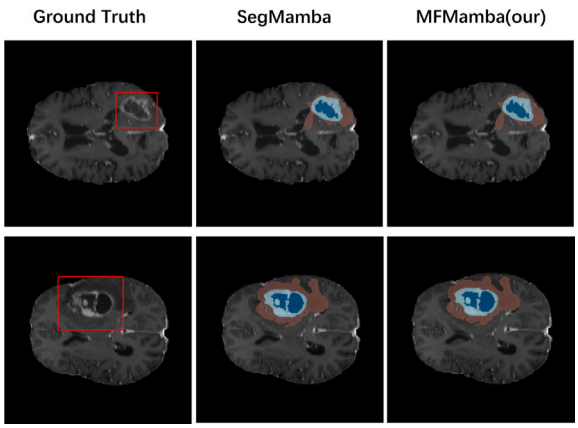


Fig. 2. Qualitative comparison on the T1c modality from BraTS 2023, showing ground truth, SegMamba, and our MFMamba predictions. Differences in tumor subregion segmentation are visually highlighted.

4.2. Implementation details

The proposed method was implemented based on the PyTorch framework and executed on a single NVIDIA RTX 4090 GPU (24 GB). Training was performed on randomly sampled 3D patches of size 128^3, with a batch size of 2 for both the segmentation and MAE networks. One epoch is defined as 250 iterations (mini-batches), following the patch-based sampling strategy commonly used in BraTS segmentation tasks, rather than a full pass over the training set. This ensures balanced coverage of foreground and background regions through oversampling.

We used stochastic gradient descent (SGD) for optimization. To train the segmentation network, we used an initial learning rate of 1×10^{-2} with momentum of 0.99 and Nesterov acceleration, and the MAE pretraining network was optimized with a learning rate of 3×10^{-4}, weight decay of 1×10^{-5}, and the same momentum and Nesterov parameters. Training was scheduled for up to 1000 epochs (250k iterations), although convergence is typically observed around epoch 150. A hybrid Dice + cross-entropy loss was applied to segmentation, whereas mean squared error (MSE) was used for MAE reconstruction.

As for the model details, the specific parameters for the MAE pretraining model and the Mamba-based segmentation model are provided in Tables 4 and 5, respectively. In the Mamba-based segmentation model, the Encoder and MambaEncoder modules were trained alternately. The MambaEncoder includes the Gated Fusion Feature (GFF) module, which is crucial for modality fusion during processing.

4.3. Evaluation metrics

We evaluate the segmentation quality using the Dice Similarity Coefficient (DSC) and the 95th percentile Hausdorff Distance (HD95) on the tumor subregions: whole tumor (WT), tumor core (TC), and enhancing tumor (ET). Higher Dice and lower HD95 values indicate better performance. Statistical significance of improvements is assessed using paired t-tests with $p < 0.05$.

Table 1
Comparison with state-of-the-art methods on the BraTS 2023 Test set. Higher dice scores and lower HD95 values indicate better performance. The table also includes the number of parameters (Params) and FLOPs for each method.

Methods	Avg		WT		TC		ET		Params (M)	FLOPs (G)
	Dice↑	HD95↓	Dice↑	HD95↓	Dice↑	HD95↓	Dice↑	HD95↓		
SegResNet [29]	88.26	4.01	92.02	4.07	89.10	4.08	83.66	3.88	50.2	90.5
UX-Net [30]	89.69	4.81	93.13	4.56	90.03	5.68	85.91	4.19	56.4	120.3
MedNeXt [31]	88.04	4.72	92.41	4.98	87.75	4.67	83.96	4.51	52.5	95.1
UNETR [16]	87.68	5.49	92.19	6.17	86.39	5.29	84.48	5.03	69.8	112.7
SwinUNETR [17]	88.23	4.70	92.71	5.22	87.79	4.42	84.21	4.48	72.1	118.4
SwinUNETR-V2 [32]	89.39	4.51	93.35	5.01	89.65	4.41	85.17	4.41	75.3	123.5
SegMamba [3]	**91.32**	**3.56**	93.61	3.37	92.65	**3.85**	**87.71**	**3.47**	67.4	110.9
Our Method	91.04	3.67	**93.77**	**3.32**	**92.69**	3.89	86.67	3.76	72.3	116.2

Table 2
Distribution of cases across training, validation, and test sets of BraTS 2023 database.

Subset	Cases	Modalities	Annotations
Training	612	4	Yes
Validation	175	4	Yes
Internal Test	88	4	Yes
Challenge Test	376	4	Hidden

Table 3
Ablation study on BraTS 2023 test set (WT Dice). (A) modality-aware masking, (B) joint optimization, and (C) gated fusion. M1 represents the baseline model, SegMamba [3], which is the most advanced Mamba-based model. The table shows the impact of different components (A, B, C) on the model's performance, with Dice score (↑) and HD95 (↓) as evaluation metrics.

Methods	A	B	C	Dice↑	HD95↓
M1				93.61	3.37
M2	✓			87.65	5.46
M3		✓		89.33	4.63
M4	✓	✓		92.56	4.85
Full Model	✓	✓	✓	**93.77**	**3.32**

4.4. Comparison with SOTA methods

We compare our proposed MFMamba against several leading methods in 3D medical image segmentation, including SegResNet [29], UX-Net [30], MedNeXt [31], UNETR [16], SwinUNETR [32], SwinUNETR-V2 [17], and SegMamba [3]. Table 1 shows both the segmentation performance (Dice and HD95) and model complexity (Params and FLOPs) of different methods. MFMamba outperforms existing models on WT and TC, achieving a Dice score of 93.77% and 92.69%, respectively, with significant improvements over SegMamba and Mamba-based models. Specifically, the improvements over the best-performing baseline (i.e., the SegMamba) are statistically significant ($p < 0.05$) on WT and TC tumors. Examples of tumor subregion segmentation by MFMamba are shown in Fig. 2.

4.5. Ablation studies

To examine the contribution of each component in MFMamba, we perform ablation studies on the BraTS 2023 test set. We isolate the effects of the modality-aware masking (A), joint loss strategy (B), and gated fusion module (C). Table 3 summarizes the results. Removing any of the components causes performance to decline. Without modalityaware masking (A), cross-modal dependencies are underutilized, reducing segmentation accuracy. Excluding the joint optimization (B) or gated fusion (C) also leads to lower Dice scores and higher HD95 values. The full model, with all components enabled, performs best.

Table 4
Architecture and parameters of MAE Model (↑ represents the Upsample operation.)

Section	Layer type	Number	Activation
Input	–	4	–
Encoder	Conv3D	16 → 128	ReLU
Decoder	↑Conv3D	128 → 16	ReLU
Output	Conv3D	16 → 4	Softmax

Table 5
Architecture and parameters of Mamba-based segmentation model (Encoder and Mamba process in a loop. Modal fusion is performed in the Mamba layer).

Section	Layer type	Number	Activation
Input	–	1	–
Encoder	UnetrBasicBlock	48 → 384	ReLU
Mamba	Mamba, GFF	4	ReLU + Sigmoid
Decoder	UnetrUpBlock	384 → 48	ReLU
Output	UnetOutBlock	48 → 13	Softmax

5. Discussion

The proposed MFMamba framework achieves reasonably good performance on the BraTS 2023 dataset, demonstrating clear improvements in Whole Tumor (WT) and Tumor Core (TC) segmentation compared to baseline methods. With Dice scores of 93.77% for WT and 92.69% for TC. The framework has been shown to be highly effective, even under conditions where one modality is missing. The proposed MFMamba maintains robust segmentation quality despite modality loss, suggesting its resilience and adaptability. The success of MFMamba can be attributed to the integration of modality-aware masked autoencoder (MAE) pretraining and gated fusion. The MAE pretraining enables the model to reconstruct missing modalities through cross-modal learning, strengthening its ability to capture complementary features from the remaining modalities. The gated fusion module further refines the task-specific features by selectively incorporating the learned generative priors, enhancing the model's overall segmentation accuracy.

Despite these promising results, some limitations remain. The performance on Enhancing Tumor (ET) segmentation is slightly lower compared to WT and TC. A possible explanation is that during the MAE pretraining, fine-grained discriminative information may be lost, particularly for complex tumor regions. These imperfect reconstructions could lead to amplified errors when fused, causing a decrease in performance. Furthermore, failure cases emphasize this issue. In cases where tumors present heterogeneous appearances or weak contrast between modalities, the MAE occasionally generates suboptimal reconstructions, affecting the fused representations. This often results in mis-segmentation, especially along irregular tumor boundaries or in small enhancing tumor regions. These observations suggest that, while generative priors provide useful guidance, they may also propagate uncertainty in more challenging cases, indicating a need for more reliable and consistent structural priors across modalities.

Another important aspect to consider is the computational efficiency of MFMamba. While our approach introduces a modest increase in model parameters and FLOPs compared to existing methods, the added complexity may not always justify the performance gain in clinical contexts where speed and simplicity are essential. However, we show that the model is feasible for both training and inference on high-end GPUs, with training memory consumption around 16–17 GB and inference taking approximately 2–3 min for a full validation epoch of 175 cases. Although this trade-off is acceptable for research and clinical workflows focused on accuracy, future work will explore lightweight adaptations to enhance computational efficiency for broader deployment.

Future work will focus on the extension of the method to improve fusion strategies and cross-domain generalization. In particular, we will investigate efficient self-attention mechanisms to enhance long-range interactions across modalities without incurring prohibitive memory costs. Another promising direction is to evaluate the framework under domain shifts, such as data from different hospitals or scanner types, to ensure robustness and applicability in diverse clinical settings.

Finally, the generalization of MFMamba to handle missing or completely unseen modalities is a critical consideration. Unlike many baseline methods that require all modalities to be available, MFMamba leverages modality-aware MAE pretraining to handle missing modalities during inference. This makes it particularly relevant in clinical practice, where incomplete acquisitions are common. However, extending the framework to entirely new modalities or tasks outside the brain domain presents an open challenge. Future research will focus on exploring attention-based fusion strategies to improve modality interaction. Although current 3D self-attention designs are memory-intensive, efficient variants may offer a promising path to more scalable multimodal integration.

6. Conclusion

We developed MFMamba, a multimodal brain tumor segmentation framework that combines modality-aware generative pretraining with an efficient Mamba-based segmentation backbone. On the BraTS 2023 benchmark, MFMamba achieved Dice scores of 93.77% for Whole Tumor (WT) and 92.69% for Tumor Core (TC), representing improvements of up to 2.1% over competitive baselines. These gains are statistically significant ($p < 0.05$) and demonstrate the benefit of integrating generative priors into downstream segmentation tasks. The method is potentially useful for enhancing delineation consistency across challenging tumor subregions for clinical diagnosis. Our method achieves a balance between segmentation accuracy and computational efficiency. While our framework introduces a moderate increase in parameters and FLOPs compared to conventional baselines, it remains computationally feasible on a single high-end GPU, with acceptable training and inference costs. More importantly, the modality-aware design naturally equips the framework to handle missing inputs, a common issue in clinical practice.

CRediT authorship contribution statement

Yaya Huang: Writing – original draft, Formal analysis, Data curation, Conceptualization. **Litong Liu:** Writing – original draft, Formal analysis, Data curation. **Tianzhen Zhang:** Writing – original draft, Validation. **Sisi Wang:** Writing – original draft, Data curation. **Chee-Ming Ting:** Conceptualization, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The BraTS 2023 dataset used in this study is publicly available at: BraTS2023.

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